

1) Objective:

• Guide Stu

Title:- Exploratory Data Analyse (EDA): Downloading the dataset and perform cleaning of data. Data Analysis & Visualisation Numpy, Pandas, matplotlib / Seaborn, Scipy.

1) Objective:

- Guide Students to download a dataset & perform essential data preprocessing & cleaning.
- Conduct exploratory data analysis using Python tools: Numpy, Pandas, Matplotlib, Seaborn & Scipy.

2) Software & Setup Requirements:

- Install a recent version of Python.
- Preferred editor: Visual Studio Code
- Install required libraries using pip:
- pip install numpy pandas matplotlib seaborn scipy

3) Background Theory:

Students should have preliminary understanding

- EDA fundamentals - Understanding data, outliers, and distribution patterns.
- Python's EDA tools - Numpy, Pandas, Matplotlib and Scipy.

- Practical Instruction:

- Step A: Download & Load Dataset.

- Choose a dataset and place it in your working directory
- import Pandas as pd
- `df = pd.read_csv('your_dataset.csv')`

- Step B: initial Exploration

- Use:

- `df.head()`

- `df.tail()`

- `df.info()`

- `df.describe()`

- Step C: Data Cleaning

- Missing Value Detection: `df.isnull().sum()`
- Handling Missing Values: e.g. `df.dropna()`, or `df.fillna(value)`
- Removing Duplicates
- Type Conversion

- Step C: Statistical Analysis

- Use Numpy for Numerical Insiging
- import numpy as np.
- `np.mean(df['column'])`, `np.median`.
- use pandas, groupby and `agg()`;
- `df.groupby.agg`

Step E: Visualization

- Matplotlib lib & seaborn

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
import warnings
```

```
warnings.filterwarnings('ignore')
```

```
sns.histplot(df[numerical_column], kde=True)
```

```
sns.heatmap(df.corr()) annot=True; cmap='cool'
```

```
sns.scatterplot(x='col1', y='col2',
```

Step F: Interpretation & Summary

- Summarize key findings:
 - Presence of missing data or anomalies and how you address them
- Outliers or surprising distribution patterns.
- Practical implications or next steps.

5. FAQ's

- What is Data frame?
A 2D labeled data structure in pandas
- Difference between head() & tail()?
head() shows the first few rows (default 5),
while tail() shows the last few
- How to get data type details?
use df.dtypes or df.info() to inspect column types.

conclusion: -

Thus we have implemented Exploratory Data Analysis : Downloading the dataset and perform Cleaning of data Analysis & Visualization - using Numpy Pandas matplotlib / Seaborn. scrapy .